Interview Analysis

1. Main themes from the interviews

The most useful finding is that the interviews do not point to one clear feature that everyone wants. Instead, they show a set of recurring tensions around price, time, personal priorities, social influence, and shopping routines. These themes should drive the next design, rather than carrying the app forward just because it already exists.

Theme	**What was said in the interviews**	**How can we use this for our next idea**
Price often comes before health	Participants repeatedly mentioned prices, sales, half-price items, cheaper brands, weekly budgets and price per 100 g. Several people also noticed immediately that the prototype did not show prices.	Price needs to be visible when it is relevant. It should not be treated as a secondary feature.
Time changes the decision	Some participants described grocery shopping as a quick in-and-out task. When rushed, people reported choosing convenient items, nearby items or easier meals instead of the original plan. Logging extra information while shopping was also seen as annoying.	Anything used in-store should be quick and low in effort. It should not ask people to constantly enter information.
Health means different things to different people	Participants used different signals to judge food, including protein, sugar, fat, sodium, ingredients, freshness, processing and even a general feeling that one item looked healthier. Some participants said health barely affected their choice.	The product should not decide what is healthy for the user. It should support the priorities the person actually cares about.
Social influence is real, but it depends on who the person is	Participants described changing purchases because of a friend, partner, flat mate or familiar recommendation. However, some people said they would only trust recommendations from someone with similar habits, budget or constraints.	Keep the idea of trusted social influence but avoid a generic social feed.
Social comparison can motivate or make people uncomfortable	Some participants liked goals and competition. Others said comparison could feel judgmental, create insecurity or make people compare budgets and lifestyles that are not equal.	Do not build the next concept around public rankings or pressure. Social features should be optional and non-judgmental.

Lists and coordination may matter more than social media features	People already use lists, send lists to other shoppers, have partners add items, and suggest shared shopping lists during the prototype feedback.	The stronger social opportunity may be coordination between people rather than posting or sharing progress.
Timing matters	Some people wanted recipes or planning support before entering the store. Others liked a simple in-store prompt when it was immediately actionable. Several people did not want a lot of information during mid-shops.	Separate planning support from decision-point support. Do not show everything all the time.
Routine and familiarity are strong	Several participants said they repeatedly buy the same foods, brands, or products they know from family and previous experience.	The next concept should fit existing habits instead of assuming users want to discover new things in every shop.

2. What the app prototype helped us learn

Making the app and the breakdown was useful because it gave participants something concrete to react to. The important outcome is not that the app should now be improved. The reactions to the app exposed what people cared about underneath the interface.

- Some people liked friend recommendations, but only when they were relevant, cheap, easy or came from someone they trusted. [1][19]
- Several people wanted prices and clearer shopping information more than social content. [2]
- Group goals were mixed. Some liked the accountability or competition, while others saw pressure, judgement or unfair comparison. [3][20]
- Participants repeatedly returned to shopping lists, coordination, price and the question of what to buy right now. [4]
- One participant explicitly rejected the standalone app idea because existing supermarket apps already cover many shopping functions. [5]
- Studio feedback suggested the prototype could feel more like a cooking-sharing app than a shopping tool. [22]

3. Design principles for the next concept

Low effort: The product should not require constant logging or long interactions while the person is moving through the store. [6]

Price aware: Price, specials and budget should be easy to see when they matter. [7]

User-defined priorities: The product should not label food as universally healthy or unhealthy. Different users care about different things. [8][21]

Optional social input: Recommendations from trusted people can be useful, but users should not be forced into public comparison. [9]

Supports existing lists: The concept should fit the way people already plan and coordinate shopping.[10]

Useful at the right moment: Planning information may be useful before the shop. Quick comparison information may be useful at the shelf. [1]

Works for solo and shared shopping: Many users shop alone, but other people still influence what they buy before, during or after the trip. [11]

Does not create another phone task: The next concept should be standalone and usable without opening another app. [12]

4. Transition from interviews to design

The team did at least 15 interviews across two rounds, each serving a different purpose. The 1st round was aimed at broadly understanding the topic and our target audience. We asked about how a recent grocery trip went, a time a friend/housemate changed what they bought, social media's influence in food choices, health vs cheaper food trade off and the hardest part of eating well as a student. The 1st round was to get a general idea of how UQ students think and how they navigate grocery shopping. The 2nd round was more for story-driven responses and prototype feedback. The participants used an early prototype, they thought aloud and explored the app. After they answered follow-up questions on features. This round was made to try and get reactions and experiences that could help with our design decisions.

The interviews didn't point to one clear feature/design everyone wanted. Instead, they showed different struggles such as price vs convenience, the value of time, influence of social media vs what they need and how being 'healthy' isn't too much of a concern (or not on their mind). The value of the early app prototype wasn't the app itself but its the people's reaction to it since it shows what actually matters to them. Because of this, we had to find different concepts that fit the problem.

We shifted away from an app based concept based on comments/analysis from interviews and from our week 5 proposal. One participant didn't like the app idea, stating that "I think there's already too many apps out there like, you know, the Coles app, or Woolies app... this is just another generic app, compared to like specific stores... So if I go to Woolies, I might as well just use the Woolies app, where I can see like, you know, real time discounts". Furthermore, Noah's review from our proposal presentation noted the team had "focused mainly on applications and not devices," suggesting alternatives like a shopping bag or trolley-mounted scanner, or a countertop scanner/projector for the home bench. These suggest the problem wasn't a feature it was the format. This made us transition to a physical device rather than an app.

5. New concepts

Concept A: The Grocery Baton

A small standalone device that lives on the fridge at home and clips onto a trolley or basket during the shop. At home it can hold a shared list and simple household notes. In-store it becomes a minimal shopping list, budget tracker and product comparison tool.

- Shared list and household notes come from participants already sending lists, sharing groceries and adding items for partners or flat mates. [10]
- A running budget responds to the repeated importance of price. [13]
- A product comparison mode can show the criteria the shopper chooses, such as price, protein, sugar or ingredients, without calling one item healthy. [14]
- The in-store display stays simple because participants described being rushed and not wanting to log information constantly. [15]

This concept gives the team the most room to test several interview findings at once without turning back into an app. It can be made as a rough physical prop and changed very quickly between participants.

How the concept related to interviews

Prototype feature	Why it is there
Shared list	Participants already use lists, send lists to other shoppers and ask for shared list support.
Budget display	Prices, specials and weekly budgets appeared repeatedly.
Prices during comparison	Multiple participants specifically wanted prices in the app prototype.
Compare mode	Participants described comparing products using price, protein, sugar, fat, ingredients or familiarity.
No universal healthy score	Definitions of health were inconsistent across participants, and some did not care about health at all.
Optional friend or household input	Social recommendations can influence purchases, but not everyone wants a feed or comparison.
Minimal in-store interaction	Participants described quick shops and disliked the idea of constant logging.
No leaderboard	Competition helped some people but made others feel judged or insecure.

Prototype idea

- A cardboard device roughly the size of a calculator.
- A fake display that can be swapped by the researcher.
- Three physical buttons: list, price and compare.
- A fake barcode scanner or scan button.
- A detachable fridge dock or simple home stand.
- A few paper household notes or meal cards.
- A running budget display.

Other concepts:

Concept B: The Budget Buzzer

A trolley-mounted device that keeps a running total. It gives a small sound or vibration when the shopper is getting close to the budget they set before the trip.

- Tests on whether live budget feedback actually changes choices.
- Directly responds to repeated comments about weekly budgets, cheaper brands and sales. [7]
- It can also test whether constant budget feedback becomes annoying or stressful.

Concept C: The Shopping Traffic Light

A small physical device with three large buttons. For example: CHEAP, QUICK and MY PREFERENCE. The shopper can press the priority that matters most at that moment.

- Tests on whether priorities change across the same shopping trip.
- Responds to interview evidence that people shift between price, time, familiarity and nutrition depending on the situation. [16]
- Could be prototyped with cardboard buttons before any electronics are built.

Concept D: Friend Recipe Vending Machine

A physical kiosk at the entrance of a store or student accommodation. It offers simple recipe cards with price, time, servings and ingredients, then prints a short shopping strip.

- Keeps the useful part of friend or peer recipe inspiration.
- Adds price and effort because participants repeatedly asked for practical information. [17]
- Avoids social feed and does not require the shopper to post anything.

Concept E: Two-Person Shopping Pole

A deliberately exaggerated shared-shopping device attached to a trolley. A shared list can be divided between two people so each person can see which section or items the other person is covering.

- Responds to participants who shop together but split up inside the store.
- Directly explore the studio feedback asking for shared shopping lists and group coordination. [18][23]
- Useful even if the final design ends up looking completely different.

Concept F: The Anti-Healthy Scale

A physical comparison station where two products can be placed side by side. The shopper moves sliders such as price, protein, sugar, convenience or familiarity. The display helps compare the two products based on those selected priorities.

- Direct response to the finding that healthy meant different things to different participants. [8][24]

- The concept deliberately avoids a single HEALTHY score.
- Makes the user choose what matters instead of the system deciding for them.

6. Shadowing

The interviews tell us what participants remember or believe they do. The next stage should observe what happens while they are actually shopping. The aim is to see where the interview themes appear in real behaviour and where the behaviour contradicts what people say.

How we should structure it

Stage	What happens
Part 1: Before entering	Short questions about the trip, list, budget, time pressure and who they are buying for.
Part 2: Normal shopping	Observe the first part of the shop with as little interruption as possible.
Part 3: Introduce the physical prototype	Give the participant the Grocery Baton or another low-fidelity concept and let them decide when to use it.
Part 4: Post-shop discussion	Ask about actual decisions that happened during the trip, which prototype parts were useful and what should change next.

Before entering the store: short questions

- What are you here to buy today?
- Do you have a list? Can I see how you are using it?
- Are you shopping for yourself or anyone else?
- Do you have a rough budget today?
- Are you in any kind of rush?
- Is there anything you are specifically prioritising today, such as price, convenience, a particular nutrition goal, something familiar, or something else?

Pretty much the same questions as the interviews but shortened

What things should we check for when theyre shopping

- Uses a pre-existing shopping list
- Goes directly to a planned aisle
- Wanders or takes a detour
- Checks a special or deal display
- Look at the shelf price
- Checks unit price or price per 100 g
- Changes brand because of price or sale
- Choosing a familiar product without comparing
- Picks up two similar products to compare
- Reads the front of the package
- Turns the package around to check nutrition or ingredients
- Picks something up and then puts it back
- Substitutes something from the original plan
- Choosing convenience because of time pressure

- Responds to a discount tag or promotional label
- Mentions something that is already at home
- Mentions another person's preference
- Talks to or asks another person for input
- Receives a recommendation from a shopping companion
- Splits up from a shopping companion and coordinates the shop
- Pauses or appears uncertain
- Makes a very fast choice without visible comparison
- Gives up on an aisle or product because finding it takes too long
- Make an unplanned purchase

Simple timing and count measures

- Time from entering the store to the first item.
- Total shopping time.
- Number of product comparisons.
- Number of substitutions from the original plan.
- Number of unplanned purchases.
- Number of times a nutrition or ingredient label is checked.
- Number of times the participant refers to another person or household need.

Questions about if they make a noticeable decision

- What made you choose that one?
- I noticed you picked the other one up first. What changed?
- What were you checking on the back?
- Why did you skip that aisle?
- Was that already on your list?
- What made that special worth changing your plan for?
- Would you normally stop this long here if I was not with you?
- You mentioned your housemate or partner there. What would normally happen if they were not shopping with you?

Post-shop questions

- What mattered most during that shop?
- Was there anything you did today that surprised you?
- Where did you spend most of most of your time deciding?
- What information did you repeatedly look for?
- Were there times you knew what you wanted immediately?
- Did anything make you change something that was on your original list?
- Which part of the prototype did you use?
- Which part did you ignore?
- Was there anything it showed you that you already knew?
- Did it ever slow you down?
- Did it make any decision easier?
- Did it make any more annoying decision?
- Did the budget display affect what you bought, or did you ignore it?
- When comparing two products, did the information shown match what you actually cared about?
- Would you want another person's recommendation to appear here? If yes, whose? If no, why not?
- Was there anything here that felt judgmental?

- If you could physically remove one feature from this device, which one would you remove?
- If you could replace that feature with something you needed during today's shop, what would go there?

7. Explaining how shadowing could help us

Contextual

The interviews show that grocery decisions depend on the setting: being rushed, seeing a sale, finding an item near the checkout, shopping with another person, or noticing what is physically available. Contextual inquiry helps study the task where it actually happens, instead of relying only on what people remember later.

Real-life Situations

The interviews suggest that plans can change once the person is in the store. A participant may have a list but still switch brands because of a sale, or replace a planned meal because time is running out. Shadowing helps capture how the actual situation shapes the choice.

Ecological validity

An interview cannot reproduce crowded aisles, physical shelf placement, a limited amount of time, carrying a basket, walking back to another aisle, or trying to coordinate with a partner. Observing shopping in the real environment gives the team better evidence about whether a concept fits the task.

What next

The next stage can compare three different sources of evidence:

1. Interview: what participants say they do.
2. Shadowing: what participants do.
3. Prototype probe: what participants do when a possible intervention is introduced.

The useful result is often the difference between these three. For example, a participant may say price is always the priority, then choose a more expensive familiar product in-store and ignore the budget display. That contradiction becomes a new design insight. [4]

8. Research reflection and limitations

The first interview round is still useful, but it is worth being open about its limitations. Some prototype questions were leading. In a few interviews, the prototype was explained in a positive way before the participant had fully reacted to it, or the interviewer suggested ideas such as accountability and healthier eating while asking the question.

This is another reason to move toward contextual observation. The next round can rely more on neutral prompts and observed behaviour, rather than repeatedly asking participants whether a feature sounds useful.

Most of the participants who were approached for shadowing declined to take part. They cited feeling watched/loss of privacy, self-conscious, discomfort with an observer present for something as personal as their budget and food choices or not having time for it due to assignments/personal reasons. This is not a recruitment problem rather than a downside of shadowing as a method. Being observed can

change how people shop now, and grocery shopping isn’t a task people expect to have an audience for. Combined with the extra time commitment compared to an interview, this shows that having low participation isn’t a fluke nor bad luck it could be a sign to change how we approach our methods.

Possible report wording

Because the prototype interviews relied on self-report and some questioning may have shaped participants' responses, the next research phase uses contextual observation to examine naturally occurring shopping behaviour with less researcher prompting.

9. Future Prototype Testing

This interview round was different from the earlier rounds. Around half of the interviews were shown a physical prototype of the cart unit alongside three screens: shopping list, comparison and alerts/housemate suggestions screen. While the other half was shown a prototype made from html which showed not only the cart unit but the home unit as well. Not every participant was shown the same version of the concept, mainly because the team only had a limited number of physical cart units built by the time interviews were scheduled. With just one or two working props available, the team couldn't run physical prototype sessions with every participant back-to-back, so some interviews used the HTML/digital walkthrough of the same concept instead, simply to keep the interview schedule moving without participants waiting on prop availability.

The reactions were more positive than the earlier rounds. Three of four app participants from Aman’s interviews (who used the html version) said they'd use the app without hesitation: "Oh yeah, for sure" (Interview 1), "Oh, 100% I would" (Interview 2), "Absolutely... I do even pay for a subscription" (Interview 3). Only Interview 4 was conditional.

Friend-sharing was liked in nearly all this round, "great to connect through friends" (Interview 1) and "very motivating" (Interview 3) were used to describe the screen. The only concern that was brought up was that Interview 1 wanted real ingredient/brand names, and Interview 4 noted recommendations could be "a tease to your eyes" when tightly budgeted.

Group goals were positive but budget-sensitive, echoing earlier findings. Interview 4 said it "could be stressful for one person because if he's getting less pay” the same insecurity pattern seen in previous interviews (mainly in the 2nd round of interviews)

A few small requests reinforce decisions already in the building: Interview 1 wanted ingredient/brand specificity, echoing Kenan's first participant, who asked for "specific brand" on ingredients like chicken. Interview 1 also suggested a due date on group goals; Interview 2's only critique was cosmetic formatting.

Overall, this round supports several existing decisions. The repeated brand-specificity request justifies using it for later prototypes. Affordability is still a big concern which justifies avoiding a “healthier/worse” verdict in the comparison screen. matching Kenan's second participant, who liked that the compare screen "doesn't tell me which one is 'healthier' or 'better'" and separately warned that a slow scan-and-compare flow would get skipped "if scanning two items takes more than like 10 seconds each, I'm just going to grab the cheaper one". Friend recommendations being valued only as pre-trip inspiration, never as an in-store feed, supports dropping the social feed from the physical device. And the value placed on simplicity "very minimal, which is good", Interview 2 matches Kenan's first participant's own speed concern: "if it adds any real delay to a normal trip I'll skip it entirely."

10. Evidence

1. "I think I'd want to see it before I go in the store… just to see what other people are eating and if it looks good" (3rd Interview) vs. "I don't really care about Sarah's chicken burrito bowl, to be honest" (3.5 Interview) recommendations landed differently depending on whether they came from someone trusted.
2. "I'm not seeing any price because for a grocery app, I'd like to see… how much the products are" (3.5 Interview); "If we added prices next to the ingredient lists, I think that would sort of save me time" (3rd Interview) both participants raised this unprompted.
3. "it's essentially telling me that I'm fat… it could cause body dysmorphia" (3.5 Interview) vs. "I think it's mostly positive than it is negative… just to kind of keep you on track" (3rd Interview) though the same, more positive participant still described "a little bit of insecurity" once he imagined prices next to a friend's shop.
4. "I don't really depend on my list at all… I just depend on my mood" (DECO3500 Interview 1); "even though I have a list, I don't always stick completely to it" (Anujitha Interview 2).
5. "I think there's already too many apps out there like… the Coles app, or Woolies app… this is just another generic app" (3.5 Interview).
6. "I'm usually a pretty quick in-and-out person… I don't try to like wander around" (3rd Interview); confirmed on the physical prototype: "if it adds any real delay to a normal trip I'll skip it entirely" (physical prototype interview).
7. "I prioritize price over the quality of what I eat… 99 out of 100 times" (DECO3500 Interview 2); confirmed physically: "sometimes I forget my budget so seeing what's remaining will really help" (physical prototype interview).
8. "I try to look at the highest nutrient value for my personal use which is protein and less fat" (3rd Interview) a personal criterion that would not suit every shopper.
9. "it's easier to try something when someone you know has already recommended it" (Anujitha Interview 2) vs. "I don't really care about Sarah's chicken burrito bowl" when the recommendation came from an app rather than someone trusted (3.5 Interview).
10. "I already have a shared list via SMS… having it on the cart unit will be really helpful, I don't have to remember what my roommate texted me a few days ago" (physical prototype interview).
11. "My housemates are quite good at finding cheap alternatives… every time I go shopping they always have to remind me to choose the more affordable ones" (Second Interview Kenan).
12. "So if I go to Woolies, I might as well just use the Woolies app" (3.5 Interview).
13. "Watching the running total update is satisfying, kind of like a mini dopamine hit… stops me from doing that thing where I get to the checkout and I'm $15 over what I meant to spend" (physical prototype interview).
14. the compare screen "doesn't tell me which one is 'healthier' or 'better'" (physical prototype interview).
15. "Am I standing in the aisle scanning two cans of peas while people wait to get past me? If scanning two items takes more than like 10 seconds each, I'm just going to grab the cheaper one and move on" (physical prototype interview).
16. the same participant chose regular olive oil over extra virgin "even though it was the more expensive option" for health reasons, but said "most of the time I choose the more convenient one when I'm low on money" (Second Interview Kenan) priorities shifted within one shopper.
17. one participant likened the feature to "Strava… you can see other people's… running stats," but wanted it pre-trip as inspiration rather than an in-store nudge (3rd Interview).
18. "when we plan to cook something together… we'll buy ingredients that everyone can share" (Anujitha Interview 1).

19. The friend-sharing feature landed very differently elsewhere: one participant said he would "absolutely" use the app and would "even pay a subscription… I'd especially enjoy the friends… friend system" (Prototype Interview 3), and a studio team called it "a great invention" (Team 16, studio interviews).
20. Group goals were also read as a positive, not a punishment, by at least one participant: "it will also kind of pressure the… users like a peer pressure to actually be healthy, so it's actually kind of good for them" (Prototype Interview 3) a direct counterpoint to the insecurity/body dysmorphia reaction elsewhere.
21. Reinforced by studio feedback: "doesn't really care if an option is healthy" (Team 16, studio interviews); one participant defined healthy personally as "using all natural ingredients" rather than any fixed score (Prototype Interview 3).
22. Corroborated directly by a studio team's own read of the concept: "their main concern is that it's more like a cooking share app, not really focused on the shopping itself" (Team 3, studio interviews).
23. A studio team suggested a more specific version of this: "there could be a shared shopping list that multiple people can use when shopping at the same time" (Team 16, studio interviews).
24. A studio team proposed a closely related feature: "something that can help people understand what is health and what is unhealthy… a sorting system… [showing] the price gap between healthy marketed food and less healthy food" (Team 3, studio interviews).